

Richard Liu

guanzhi@uchicago.edu | [Github](#) | [Website](#)

Education

University of Chicago 4th year PhD Student, Computer Science Advisor: Rana Hanocka	2022 – Expected 2027
---	----------------------

University of Chicago B.A. Economics, B.S. Mathematics (GPA 3.73)	2015 – 2019
---	-------------

Publications

Research Interests: 3D graphics and geometry processing from 2D priors, 3D generation

Deep Feature Deformation Weights	2026
---	------

Richard Liu, Itai Lang, Rana Hanocka
CVPR 2026 Highlight ([Paper](#), [Website](#), [Code](#), [Video](#))

WIR3D: Semantic and Geometric-Aware 3D Shape Abstraction	2025
---	------

Richard Liu, Daniel Fu, Noah Tan, Itai Lang, Rana Hanocka
ICCV 2025 Oral ([Paper](#), [Website](#), [Code](#))

TEDi: Temporally-Entangled Diffusion for Long-Term Motion Synthesis	2024
--	------

Jack Zhang, Richard Liu, Noam Aigerman, Vladimir G. Kim, Rana Hanocka
SIGGRAPH North America 2024 ([Paper](#), [Website](#), [Code](#))

Towards Multimodal Interaction with AI-Infused Shape-Changing Interfaces	2024
---	------

Chenfeng Gao, Wanli Qian, Richard Liu, Rana Hanocka, Ken Nakagaki
UIST Poster 2024 ([Paper](#), [Video](#))

HyperFields: Towards Zero-Shot Generation of NeRFs from Text	2024
---	------

Sudarshan Babu*, Richard Liu*, Avery Zhou*, Michael Maire, Gregory Shakhnarovich, Rana Hanocka
ICML 2024 ([Paper](#), [Website](#), [Code](#))

DA Wand: Distortion-Aware Selection using Neural Mesh Parameterization	2022
---	------

Richard Liu, Noam Aigerman, Vladimir G. Kim, Rana Hanocka
CVPR 2023 ([Paper](#), [Website](#), [Code](#), [Talk](#))

Text2Mesh: Text-Driven Neural Stylization for Meshes	2021
---	------

Oscar Michel*, Roi Bar-On*, Richard Liu*, Sagie Benaim, Rana Hanocka
CVPR 2022 Oral ([Paper](#), [Website](#), [Code](#))

Designing Online Advertisements via Bandit and Reinforcement Learning	2021
--	------

Richard Liu, Yusuke Narita, Kohei Yata
(Peer-Reviewed) ICML 2021 RL4RealLife Workshop ([Paper](#))

Invited Talks

Symposium on Geometry Processing	2026
---	------

“Geometry Processing from 2D Image Priors”
University of Bern

Summer Geometry Institute	2024-2026
----------------------------------	-----------

“Introduction to Mesh Parameterization”
MIT

Geometric Computation Group (Guibas Lab)	2023
---	------

“Neural Mesh Parameterization: DA-Wand and Beyond”
Stanford

Professional Activities

Workshop Organizer 2025

[Great Lakes Graphics](#)

University of Chicago

Reviewer

CVPR (2023-26), SIGGRAPH North America (2023-26), SIGGRAPH Asia (2023-26), ICCV/ECCV (2023-26), Eurographics (2023), 3DV (2025-26), CAD (2023, 2024), CGF (2022), TVCG (2025)

[DA Wand Blender Extension](#)

2025

Implemented and released DA Wand trained model as a Blender extension ([extension link](#))

SGI Project Leader

2022

Summer Geometry Institute

Replicability Stamp Reviewer (GRSI)

2022, 2025

Teaching Assistant/Guest Lecturer

2022-26

CMSC 23700: Introduction to Computer Graphics

Awards

Eckhardt Graduate Fellowship

2022 - Ongoing

ITA Tech Challenge

Top 150

Dean's List

2015-2019

University Scholar Fellowship

2015-2019

Work Experience

Founding Research Engineer

2026-Present

Thrixel

University of Chicago 3DL

2021 - Present

Rana Hanocka

PhD Student

PhD Intern

Summer 2023

Adobe Research

Yale Tobin Center for Economic Policy

2020 - 2021

John Eric Humphries and Yusuke Narita

Predoctoral Fellow

Booth School of Business

2020 – 2020

Anita Rao

Research Professional

Voleon Capital Management

2019 – 2020

Operations Associate